Soumiya Ayyappan_W7_D2

Code and its output:

Step1:

Code:

```java
import java.util.*;
public class CollectionDemo{
    static void introduction(){
        System.out.println("===Introduction to Collection===");
        ArrayList<String>fruits=new ArrayList<>();
        fruits.add("Apple");
        fruits.add("Banana");
        fruits.add("Mango");
        System.out.println(fruits);
    }
```

Output:

```
===Introduction to Collection===
[Apple, Banana, Mango]
```

Step 2:

Code:

```java
static void listDemo(){
    System.out.println("\n==List Demo===");
    List<Integer>arrayList=new ArrayList<>();
    List<Integer>linkedList=new LinkedList<>();
    arrayList.add(10);
    arrayList.add(20);
    arrayList.add(30);
    linkedList.add(100);
    linkedList.add(200);
    linkedList.add(300);
    arrayList.remove(1);
    linkedList.remove(0);
    System.out.println("ArrayList:"+arrayList);
    System.out.println("LinkedList:"+linkedList);
    System.out.println("First element:"+arrayList.get(0));
}
```

Output:

```
==List Demo===
ArrayList:[10, 30]
LinkedList:[200, 300]
First element:10
```

Step 3:

Code:

```
static <T> void printArray(T[]array){
    for(T item:array){
        System.out.print(item+"");
    }
    System.out.println();
}
static void genericDemo(){
    System.out.println("\n===Generics Demo===");
    Integer[] numbers={1,2,3,4};
    String[] names={"Alice","Bob","Charlie"};
    printArray(numbers);
    printArray(names);
}
```

Output:

```
===Generics Demo===
1234
AliceBobCharlie
```

Step 4:

Code:

```
static void setDemo(){
    System.out.println("\n===Set Demo===");
    HashSet<String>names=new HashSet<>();
    names.add("Alice");
    names.add("Bob");
    names.add("Charlie");
    names.add("Alice");
    System.out.println("HashSet:"+names);
    TreeSet<String>sorted=new TreeSet<>(names);
    System.out.println("TreeSet:"+sorted);
}
```

Output:

```
===Set Demo===
HashSet:[Bob, Alice, Charlie]
TreeSet:[Alice, Bob, Charlie]
```

Step 5:

Code:

```
    static void mapDemo(){
        System.out.println("\n===MapDemo===");
        HashMap<String,Integer>marks=new HashMap<>();
        marks.put("Deva",95);
        marks.put("Rahul",88);
        marks.put("Priya",91);
        marks.put("Rahul",90);
        for(Map.Entry<String,Integer>entry:marks.entrySet()){
            System.out.println(entry.getKey()+":"+entry.getValue());
        }
    }
    public static void main(String[] args){
        introduction();
        listDemo();
        genericDemo();
        setDemo();
        mapDemo();
    }
}
```

Output:

```
===MapDemo===
Rahul:90
Priya:91
Deva:95
```